



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknološi ya Tshedimošo

Study Guide

School of Information Technology
Department of Computer Science

Imperative Programming COS 132

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Last Revision: Version 1.3 (22 April 2024)

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1 Overview

1.1 Description

This module introduces imperative computer programming, which is a fundamental building block of computer science. The process of constructing a program for solving a given problem, of editing it, compiling (both manually and automatically), running and debugging it, is covered from the beginning. The aim is to master the elements of a programming language and be able to put them together in order to construct programs using types, control structures, arrays, functions and libraries. An introduction to object orientation will be given. After completing this module, the student should understand the fundamental elements of a program, the importance of good program design and user-friendly interfaces. Students should be able to conduct basic program analysis and write complete elementary programs.

1.2 Educational Approach

The module introduces the student to programming skills so that they can design and implement solutions to software problems in the imperative programming style. The C++ programming language is used as vehicle to develop these skills. A student who assimilates the material taught in the module, will have enough basic knowledge to learn other programming languages, such as MatLab, SAS, Java and C# that may be encountered later.

Programming constructs and techniques will be presented during the lecture period. Quizzes will be made available for students to access their learning of the concepts presented in the previous couple of lectures. Tutorial sessions will be conducted to provide students with the tools to setup their programming environments on their personal computers. Tutorial sessions later in the semester will take the form of exercise sessions which will enable students to develop the necessary analytical and problem solving skills to design and implement a program solution based on a given set of requirements. The practical assignments will enable students to hone their problem solving, design and programming skills. The quizzes and practical assignments form part of the formative assessment for the module. Summative assessment, in the form of semester tests and the final examination which will test the student's understanding of the module content and the application of this content to posed problems.

1.3 Credits and Notional Hours

COS132 constitutes 16 module credits. This means that it is expected that the average student should devote at least 160 hours to the module during the course of the semester. There are 15 weeks in a semester. Thus a student should spend an average of 10 hours 40 minutes per week on this module. Three of these hours will be spent in lectures per week, one hour in a tutorial session and another three working on practical assignments in the practical sessions. This will leave 3 hours and 40min to prepare and reflect on the module content per week. 40min of this time may be assigned to an online Quiz.

1.4 Related Modules

- COS 110 - Follows on directly from COS 132 and requires a good working knowledge of the fundamental topics from imperative programming to be understood. These topics are applied to the Object-oriented paradigm.

- COS 212 - Data Structures and Algorithms
- COS 214 - Builds on the understanding of the C++ programming language and considers the design of programs using reusable constructs.
- COS 221 - Assumes knowledge of certain aspects covered within this course.

2 Study Units and Outcomes

2.1 Career Outcomes

The module provides the student with the fundamentals of imperative programming and the skills to be able to solve a basic problem. The module forms the foundation for students continuing with computer science studies. Students who do not continue with computer science will have an appreciation of the requirements and the skills required to design and code software solutions.

2.2 Study Units

The module is divided into eleven (11) study units. Each study unit will comprise a number of themes and themes are further divided into topics (with subtopics). Themes may cut across multiple study units.

The module will focus on the following themes defined across 11 Study Units. Each of these themes are further divided into topics. The topics of the themes are interrelated. Themes and topics therefore build onto one another and therefore a theme may not be presented in its entirety before the next theme is begun.

- Fundamental programming constructs
 - Basic syntax and semantics of C++
 - Variables, Expressions, Scoping
 - Assignment
 - Simple I/O
 - Control structures: Boolean Expressions, Branching and Looping
 - Functions and parameter passing
- Fundamental types and data structures
 - Primitive types
 - Arrays
 - Structures
 - Strings and string processing
- Advanced C++ features
 - Pointers
- Algorithms and problem-solving
- Programming style

2.3 Module Outcomes

After completing this module, the student should understand the fundamental elements of a program, the importance of good program design and user-friendly interfaces. Students should be able to conduct basic program analysis and write complete elementary programs.

Specific outcomes per theme are given below:

- Fundamental programming constructs
 - be able to read the syntax of C++ programs and understand what the program achieves (semantics);
 - use variables of different types and structures in coding problems to store user provided data (from the console or file) and program generated data;
 - understand and apply control structures for branching and looping;
 - identify and code units of work that are to be reused by the program into functions;
- Fundamental data structures
 - have an appreciation for the types, both simple and complex, available in C++ and know when and how to apply them;
 - manage and structure more complex data using simple data structures and algorithms;
- Advanced C++ features
 - understand what memory manage is and be able to know when to use stack and heap memory;
- Algorithms and problem-solving
 - understand the requirements for a program and be able to code a efficient program to meet those requirements;
 - identify which topics from the overall themes to apply when solving a problem;
 - design and code algorithms in functions or programs that are well written and efficient.

Imperative programming forms part of the 2013 ACM/IEEE Computer Science Curriculum Volume (CS2013)¹ in the *Programming Languages (PL)*, *Software Development Fundamentals (SDF)* and *Algorithms and Complexity (AL)* Knowledge Areas. These Knowledge Areas comprise a number of core and elective Knowledge Units. The topics presented in this module fulfil the core requirements for ACM/IEEE 2013. This module introduces the first programming language of the two languages CS2013 requires to be taught.

3 Policies and Procedures

Refer to the Departmental Guide for Academic Dishonesty and Plagiarism, Anti-discrimination and Escalation procedures, along with other matters. The Departmental guide is available on the CS portal (portal.cs.up.ac.za).

¹CS2023 is currently in Gamma version

4 Instructors

4.1 Lecturers

Name	Office no.	email	Tel no.	Comment
Linda Marshall	IT 4-28	lmarshall@cs.up.ac.za	+27 (0)12 420 3624 or +27 (0)84 828 6042	Module coordinator
Mr Sheunesu Makura	IT 4-30	makura.sm@up.ac.za	+27 (0)12 420 4116	Lecturer
Ms Seani Rananga	IT 4-29	seani.rananga@up.ac.za	+27 (0)12 429 3121	Lecturer
Mr Tlou Lebelo		u15209190@tuks.co.za		Assistant Lecturer
Ms Inge Odendaal		u21689432@tuks.co.za		Assistant Lecturer
Mr Mark Botros		u21448109@tuks.co.za		Assistant Lecturer

4.2 Tutors

The tutors assigned to the module will be available in the Informatorium during the practical sessions to answer questions about the practical assignments. The names of the tutors per lab session will be available on clickUP.

5 Organisation

Students are required to attend lectures, practical sessions, tutorial sessions and scheduled consultation times.

The module follows a **semester mark (SM) – exam mark (EM) – final mark (FM)** model. It is **imperative that a student keeps up to date**. Students that apply a continuous work ethic are more likely to succeed than students that work sporadically and tend to work in bursts before crucial deadlines, such as semester tests and the examination. Therefore, you are advised to continuously make use of the available opportunities, such as tutorials and quizzes, to make sure that you understand the work before you write your semester tests or exams. It is hard work to stay up to date, but much harder work to catch up.

In the following sections the support that is given to assist you to stay up to date is described in more detail.

5.1 Mode of Instruction

COS132 will make use of an on-campus mode of instruction for all sessions. There may be times when it will be necessary to conduct online sessions and these will be communicated on the clickUP page of the module.

5.2 Mode of Communication

clickUP (clickup.up.ac.za) will be used as the main avenue of communication for the module. Study Unit notes and Lecture Slides will be uploaded to module page on clickUP. Class Collaborate on clickUP will be used for online sessions if needed. clickUP announcements will be used to communicate organisational aspects of the module (such as the weekly schedule). The clickUP Message functionality will be used for bi-directional communication between the COS132 staff and the students.

The CS Portal, available at <https://portal.cs.up.ac.za>, will be used for requirements for which there are no tools in clickUP. Please refer to announcements on clickUP to know when you need to use the CS Portal.

Practical assignments will be submitted to an automatic marker (automarker) for evaluation. The automarker is affectionately referred to as Fitchfork and is available at ff.cs.up.ac.za.

5.3 Lecture Schedule

You are advised to prepare for lectures. To assist you, a lecture schedule will be posted on a weekly basis on clickUP in the content area. Lectures will be presented in the lecture room as per the university timetable and will be on-campus. If it is necessary for a lecture to be presented online, an announcement will be posted on the module clickUP page and a Class Collaborate session on clickUP opened.

5.4 clickUP communication tools

clickUP provides three types of communication tools. The first is visible on the Content page of the module. The Content page will be used to provide lecture schedules, notes, practical assignment specifications, links to quizzes and the like. The Announcement tool will be used to push important announcements to all students. It is recommended that students should visit clickUP regularly, not to miss anything important. Not all announcements will be emailed automatically. Messages, the final communication tool, provide an opportunity for students to respond. It is up to you to visit the clickUP page for content updates, messages and announcements to make sure you are up to date.

Note that the Department reserves the right to deviate from planned schedules under circumstances which cannot yet be foreseen, and that some instructions on clickUP might be mandatory.

5.5 Practical sessions

Each student must timetable a practical session in which to work on the practical assignments that form part of the module assessment. Teaching assistants will be available in these sessions to help with questions that may arise from the practical assignment.

5.6 Communication with lecturers

You are encouraged to ask questions during lectures.

If you have a problem that cannot be addressed before, during or after a lecture make an appointment with a lecturer to discuss the problem. You are advised to send an **email** with enough information as to what your problem might be to request an appointment. Sometimes a simple email will suffice in answering a question if enough information is provided. Make sure you send the email from your @tuks.co.za account. Begin the subject of the email with **COS132 - <short description of your email>** and to include your name in the body of email. Depending on the nature of your email, it may take up to 4 to 7 days to answer the email. If the email is applicable to the class in general, an announcement may be posted on clickUP in answer to your email.

5.7 Summary

The activity diagram given in Figure 1 provides a summary of the procedures to be followed should you require assistance in the module.

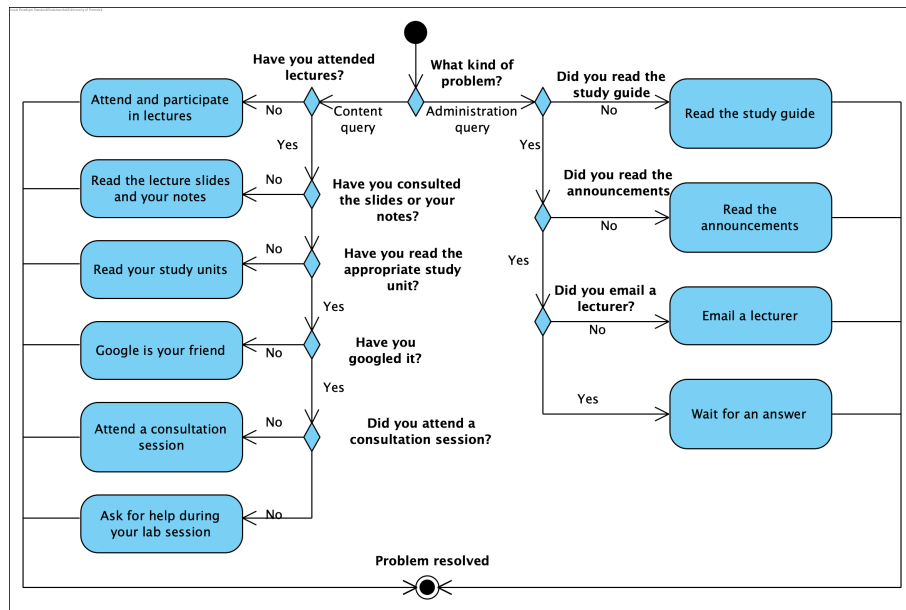


Figure 1: Organisational diagram (UML Activity diagram)

6 Study Material

6.1 Study Unit notes

Study Unit notes will be uploaded to clickUP. These notes form the basis for what is to be covered in the lectures following the publication of the notes. Make sure you read the notes and are comfortable with their content.

6.2 Software

An online compiler will be used for the first few practical assignments. This is to give you time to setup your home environment with a C++ compiler. Note, C++ has been installed in the Informatorium on Linux. During the practical sessions you will be required to make use of C++ on Linux.

The online compiler that will be used in this module is available from:

https://www.onlinegdb.com/online_c++_compiler. Details of the C++ compiler that can be installed on your own computer will be given during the allocated tutorial session.

7 Assessment

As stated in Section 5, the assessment follows the Semester Marks – Exam Mark – Final Mark **SM – EM – FM** model where SM and EM include one or more assessment events. The semester mark is made up of semesters tests, quizzes and practical assignments. The exam mark takes the form of a written examination paper which is written during the July examination period.

Each of the assessment events that make up the SM and EM constituent parts of the assessment model will be described in the sections that follow.

7.1 Semester Tests

Semester tests (STs) contribute towards the summative assessment component of the semester mark. There are two semester tests during the semester (ST1 and ST2). In both cases these tests are scheduled close to the Engineering test week. STs are scheduled centrally and the dates are available on the student portal.

If a student fails to write a ST due to illness, a valid medical certificate must be submitted by email to the module coordinator. The original copy of the medical certificate must be submitted to the Computer Science reception as per the details provided in the Departmental Guide. One aegrotat semester test opportunity will be arranged towards the end of the semester. This will replace the semester test for which you submitted a valid medical certificate.

Students with disabilities who need support and/or extra time should confirm special arrangements a week before the semester takes place. Send an e-mail to the module coordinator to request arrangements.

7.2 Quizzes

Quizzes have been scheduled throughout the semester. These will be available on a Friday and will be conducted online on clickUP. Quizzes form part of the formative assessment

component of the module. This means that you should use the quiz as a means to gauge your performance in the module and whether you have a basic understanding of the contents of the module. Using the quizzes in this way will help you prepare for the practical assignments and the semester tests and ultimately the examination on a continuous basis.

Quizzes will typically be released on Fridays. There are no aegrotat opportunities for quizzes.

7.3 Practical Assignments

Practical assignments will be presented during the semester and are regarded as the rest of formative assessment component of the module. Practical assignments are designed to provide each student with the opportunity to determine whether they have mastered the skills required to successfully complete the module. Assignment specifications will be released on clickUP and the automatic marker system Fitchfork. Submissions need to be made via Fitchfork.

Deadlines will be applied strictly. **No late submissions will be accepted.** There will also be no aegrotat assignments.

7.4 Examination

An examination is scheduled for the June examination period. The scope of this examination will be the entire syllabus for COS132. A subminimum of 40% is required in the examination to be eligible to pass the module.

Please take note of the examination rules, as provided for in the general rules and regulations of the University of Pretoria, under section G.12.

A valid medical certificate must be submitted via the student portal within 3 working days of the examination date to be able to qualify for an aegrotat exam. Please also email the medical certificate to the module co-ordinator after uploading it to the student portal.

A supplementary exam will be granted to students with a final mark between 40% and 49% inclusive. Details about the date and venue of the supplementary examination will be available on the student portal. The scope of the supplementary examination will be the same scope as for the examination.

Note that no special arrangements are made for students who are admitted to the supplementary exam but are unable to write it. If a student fails to write, the final mark that was obtained when admitted to the supplementary, will remain unchanged. If you were ill for the examination and are eligible to write a special/aegrotat examination, there is no supplementary examination if you fail to obtain the subminimum in the examination or pass the module.

The following regulations hold for supplementary exams:

- To pass the course, if admitted to the supplementary exam, a student must obtain at least 50% in the supplementary exam.
- The final mark for a student who was admitted to the supplementary exam is the mark he/she obtains in the supplementary exam.
- A maximum of 50% final mark may be awarded.

7.5 Mark Allocation

Students will be evaluated by means of semester tests, quizzes, practical assignments and the examination. The final mark for the module will be calculated as follows:

7.5.1 Semester mark (SM)

Evaluation Event	Percentage	Comment
Semester tests		
– ST1	20% or 30%	Written around Test Week 1
– ST2	20% or 30%	Written around Test Week 2
Quizzes	20%	
Practical assignments	30%	
TOTAL	100%	

Note, the best of ST1 and ST2 will count 30% towards the semester mark. The other will count 20% towards the semester mark.

7.5.2 Examination mark (EM)

The examination written in June will be converted to a mark out of 100.

7.5.3 Final Mark (FM)

The SM and the EM each count 50% towards the FM. That is $FM = (SM + EM) / 2$. A SM of at least 40% will provide access to the examination. An EM of at least 40% is required to be eligible to pass the module. A FM of at least 50% will result in a module pass, while a FM in the 40%'s will result in a supplementary exam.

8 Schedule

8.1 Lectures

There are four (4) lectures scheduled per week for the module in three (3) groups. For each lecture choose one of the groups. Lectures will be presented in English. These take place as follows:

Lecture	Day	Time	Venue G1	Venue G2	Venue G3
L1	Monday	10:30 11:30 13:30	Centenary 6 vd Bijl Hall	IT 2-23	IT 2-27
L2	Tuesday	07:30 13:30 14:30	Centenary 1	IT 2-23	IT 2-26
L3	Thursday	08:30 11:30 13:30	Centenary 6 3	IT 2-23	IT 2-26

Lectures will not take place during Engineering Test week and certain lectures will be cancelled during weeks with public holidays to ensure that all groups receive the same number of lectures.

Loadshedding is here to stay. On most days, it does not impact our teaching. It takes at most 10min when the switchover from Eskom to generators and vice-versa takes place for the power and projector to be restored. There are however times when projectors or generators fail and it is impossible to present a lecture in the venue. In these cases, a catchup lecture will be arranged. This requires a venue to be scheduled.

8.2 Tutorials

Lecture	Day	Time	Venue G1	Venue G2	Venue G3
Tutorial	Wednesday	10:30 11:30 13:30	Centenary 6	IT 2-23	Muller hall

Note, on the official UP timetable, L3 is scheduled on a Wednesday and the tutorial on a Thursday. Due to operational constraints, the tutorial session and L3 have been switched.

8.3 Practical Sessions

Space is available during the scheduled practical sessions to work on the practical assignment. The following are the times for the practical sessions.

Day	Time	Venue
Monday	14:30-17:20	tba
Tuesday	07:30-10:20	tba
Tuesday	10:30-13:20	tba
Thursday	14:30-17:20	tba
Friday	14:30-17:30	tba

Tutors will be available during the practical sessions to answer questions about the current practical assignment. Students are also welcome to use the space in the labs during these times to work on their practical assignments.

Lab allocations will be done once the Tutors have been appointed.

8.4 Consultation sessions

Students are invited to attend the consultation sessions where lecturers are available in the Costutorium (IT 4-22) for one-on-one discussions about the work. These times will be made available once the semester has begun.

9 Support services

Consult the departmental guide, available on the CS portal, for the full list of support services.

9.1 Safety in the evening and emergencies

- For any safety or emergency related matters, eg if you need a security officer to accompany you from your residence to campus, phone the Operational Management Centre (details at the back of your student card).
- The 24-hour, multi-disciplinary UP Crisis Line offers professional and confidential support to victims of crime in times of trauma. For assistance and immediate action, phone the UP Crisis Line on: 0800 00 64 28.
- Hatfield residence students: From 18:00 till 06:00 security officers are available to escort you (on foot) to and from your residence or campus anywhere east of the Hatfield Campus through to the Hillcrest Campus.

9.2 E-learning support

- Report a problem you experience to the Student Help Desk on your campus.
- Visit the open labs in the Informatorium Building or IT labs on your campus to report problems at the offices of the Student Help Desk.
- Approach the assistants at the help desks' campus specific (for example: adjacent to the Student Computer Laboratories in IT Building, NW2, CBT or Aldoel Building IT labs, etc).
- Call 012 420 3837.
- Email studenthelp@up.ac.za

10 Document Change Log

Date	Change
19 Feb 2024	Initial preparation and presentation of the document.
20 Feb 2024	Updated venue for Group 2. New version 1.1.
22 Feb 2024	Updated time and venue for Group 3 on a Wednesday. New version 1.2.
22 April 2024	Updated venues for Group 1. New version 1.3